QAOA Energy Landscape Explorer: A Complete Mathematical Walkthrough for MaxCut on the 4-Node Cycle Graph

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1 Introduction

The Quantum Approximate Optimization Algorithm (QAOA), introduced by Farhi, Goldstone, and Gutmann [1], is a hybrid quantum-classical algorithm designed for combinatorial optimization on near-term quantum hardware. QAOA alternates between a *cost unitary* that encodes the objective function and a *mixer unitary* that explores the solution space, producing a parametrized quantum state whose expectation value approximates the optimal cost.

This report presents a complete, self-contained mathematical walkthrough of the QAOA energy landscape for the **MaxCut problem on the 4-node cycle graph** C_4, at circuit depth $p = 1$. We proceed from the graph definition through the Ising Hamiltonian, circuit construction, energy landscape analysis, classical optimization, and finally noise modelling with zero-noise extrapolation (ZNE) error mitigation. Every derivation step is shown explicitly so the reader can reproduce the results from first principles.

2 Problem Formulation

2.1 The MaxCut Problem

Given an undirected graph $G = (V, E)$ with edge weights $w_{ij} > 0$, the **MaxCut** problem asks for a partition of V into two disjoint subsets S and $\bar{S}$ that maximises the total weight of edges crossing the partition:

$$\mathrm{MaxCut}(G) \;=\; \max_{S \subseteq V} \sum_{(i,j)\in E} w_{ij} \left[\, i \in S,\; j \in \bar{S} \text{ or } j \in S,\; i \in \bar{S} \,\right]. \tag{1}$$

Assigning Ising spin variables $z_i \in \{+1, -1\}$ to each vertex (with $z_i = +1$ if $i \in S$ and $z_i = -1$ if $i \in \bar{S}$), the cost function becomes:

$$C(\mathbf{z}) \;=\; \sum_{(i,j)\in E} w_{ij}\, \frac{1 - z_i z_j}{2}. \tag{2}$$

A cut edge contributes 1 (since $z_i z_j = -1$) while an uncut edge contributes 0 (since $z_i z_j = +1$).

2.2 The 4-Node Cycle Graph C_4

We consider the cycle graph on four vertices:

$$V = \{0,\, 1,\, 2,\, 3\}, \tag{3}$$

$$E = \{(0,1),\; (1,2),\; (2,3),\; (3,0)\}, \tag{4}$$

$$w_{ij} = 1 \quad \forall\, (i,j) \in E. \tag{5}$$

The cost function for C_4 is therefore:

$$C(\mathbf{z}) \;=\; \frac{1 - z_0 z_1}{2} + \frac{1 - z_1 z_2}{2} + \frac{1 - z_2 z_3}{2} + \frac{1 - z_3 z_0}{2}. \tag{6}$$

Enumerating all $2^4 = 16$ bitstrings, the maximum cut is:

$$C_{\max} = 4, \qquad \text{achieved by } \mathbf{z}^* \in \{0101,\; 1010\}, \tag{7}$$

where we use the binary encoding $z_i = (-1)^{x_i}$ with $x_i \in \{0, 1\}$. The minimum cut is $C_{\min} = 0$, attained by $\{0000,\; 1111\}$.

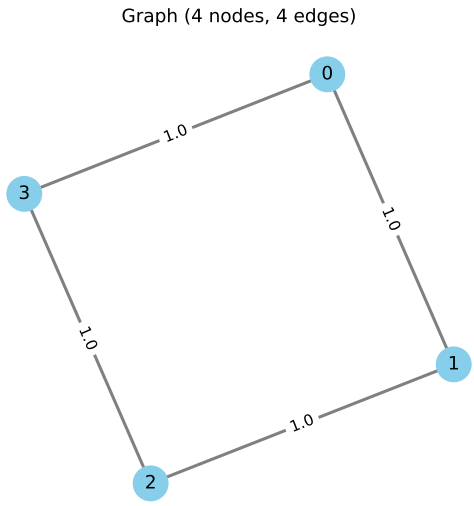

Figure 1: The 4-node cycle graph C_4 with unit edge weights.

3 Ising Hamiltonian

3.1 General Construction

Promoting the classical spins z_i to Pauli-Z operators Z_i acting on qubit i, the cost Hamiltonian is:

$$H_C \;=\; \sum_{(i,j)\in E} \frac{w_{ij}}{2}\big(I - Z_i Z_j\big), \tag{8}$$

where I is the identity operator on the full 2^n-dimensional Hilbert space. By construction, H_C is diagonal in the computational basis and its eigenvalues are the cut values:

$$H_C \left|\mathbf{x}\right\rangle = C(\mathbf{x}) \left|\mathbf{x}\right\rangle. \tag{9}$$

3.2 Explicit Form for C_4

Substituting the four edges of C_4 with $w_{ij} = 1$:

$$H_C \;=\; \frac{1}{2}\Big[(I - Z_0Z_1) + (I - Z_1Z_2) + (I - Z_2Z_3) + (I - Z_3Z_0)\Big]. \tag{10}$$

Collecting the identity terms ($4 \times \frac{1}{2} = 2$):

$$\boxed{H_C \;=\; 2I \;-\; \frac{1}{2}\big(Z_0Z_1 + Z_1Z_2 + Z_2Z_3 + Z_3Z_0\big).} \tag{11}$$

3.3 Eigenvalue Verification

Since H_C is diagonal in the computational basis, we verify selected eigenvalues. For a computational basis state $|x_0x_1x_2x_3\rangle$, each Z_i has eigenvalue $(-1)^{x_i}$.

Table 1: Eigenvalues of H_C for selected basis states of C_4.

$\lvert x_0x_1x_2x_3\rangle$	Z_0Z_1	Z_1Z_2	Z_2Z_3	Z_3Z_0	$\sum Z_iZ_j$	$C(\mathbf{x})$
$\lvert 0000\rangle$	$+1$	$+1$	$+1$	$+1$	$+4$	0
$\lvert 1111\rangle$	$+1$	$+1$	$+1$	$+1$	$+4$	0
$\lvert 0001\rangle$	$+1$	$+1$	-1	-1	0	2
$\lvert 0011\rangle$	$+1$	-1	-1	$+1$	0	2
$\lvert 0101\rangle$	-1	-1	-1	-1	-4	4
$\lvert 1010\rangle$	-1	-1	-1	-1	-4	4

Using Eq. (11), $C(\mathbf{x}) = 2 - \frac{1}{2}\sum Z_iZ_j$, which matches the expected cut values. The maximum eigenvalue is $C_{\max} = 4$ for the alternating bitstrings $|0101\rangle$ and $|1010\rangle$, confirming the classical result.

4 QAOA Circuit Construction

4.1 The QAOA Ansatz

For depth $p = 1$, the QAOA state is:

$$|\psi(\gamma,\beta)\rangle \;=\; U_B(\beta)\,U_C(\gamma)\;|+\rangle^{\otimes n}\,, \tag{12}$$

where $|+\rangle^{\otimes n} = H^{\otimes n}\,|0\rangle^{\otimes n} = \frac{1}{2^{n/2}}\sum_{\mathbf{x}\in\{0,1\}^n}|\mathbf{x}\rangle$ is the uniform superposition.

4.2 Cost Unitary $U_C(\gamma)$

The cost unitary is the time evolution under H_C:

$$U_C(\gamma) \;=\; e^{-i\gamma H_C} \;=\; \prod_{(i,j)\in E} e^{-i\gamma\frac{1}{2}(I - Z_iZ_j)}. \tag{13}$$

The product form holds because all Z_iZ_j terms commute (they are simultaneously diagonal). Expanding each factor:

$$e^{-i\gamma\frac{1}{2}(I-Z_iZ_j)} \;=\; e^{-i\gamma/2}\,e^{i\gamma/2\,Z_iZ_j}. \tag{14}$$

The global phase $e^{-i\gamma/2}$ per edge accumulates to $e^{-2i\gamma}$ for the four edges of C_4, and can be dropped since it does not affect expectation values. The non-trivial part is:

$$U_C(\gamma) \;\sim\; \prod_{(i,j)\in E} e^{i\gamma/2\,Z_iZ_j}. \tag{15}$$

4.3 Gate Decomposition: IsingZZ Gates

Each two-qubit term $e^{i\gamma/2\,Z_iZ_j}$ is an IsingZZ (or R_{ZZ}) gate. In terms of CNOT and R_Z gates:

$$e^{i\theta\,Z_iZ_j} \;=\; \mathrm{CNOT}(i,j)\;\cdot\;R_Z(2\theta)_j\;\cdot\;\mathrm{CNOT}(i,j), \tag{16}$$

where $R_Z(\phi) = e^{-i\phi Z/2}$. For our circuit, $\theta = \gamma/2$, so each edge contributes:

$$\mathrm{CNOT}(i,j) \;\to\; R_Z(\gamma)_j \;\to\; \mathrm{CNOT}(i,j). \tag{17}$$

4.4 Mixer Unitary $U_B(\beta)$

The mixer unitary applies single-qubit X-rotations:

$$U_B(\beta) \;=\; \prod_{i=0}^{n-1} e^{-i\beta X_i} \;=\; \prod_{i=0}^{n-1} R_X(2\beta)_i, \tag{18}$$

where $R_X(\phi) = e^{-i\phi X/2}$.

4.5 Circuit Diagram

The complete $p = 1$ QAOA circuit for C_4 consists of the following layers:

1. **Initialization**: Apply H (Hadamard) to all 4 qubits: $|0\rangle^{\otimes 4} \to |+\rangle^{\otimes 4}$.
2. **Cost layer** $U_C(\gamma)$: Four IsingZZ$(-\gamma)$ gates, one per edge:
   - ZZ(γ) on qubits $(0, 1)$
   - ZZ(γ) on qubits $(1, 2)$
   - ZZ(γ) on qubits $(2, 3)$
   - ZZ(γ) on qubits $(3, 0)$
3. **Mixer layer** $U_B(\beta)$: Four $R_X(2\beta)$ gates, one per qubit.
4. **Measurement**: Measure all qubits in the computational basis.

5 Energy Landscape

5.1 Energy Function

The QAOA energy (objective function) is the expectation value of H_C in the variational state:

$$E(\gamma, \beta) \;=\; \langle\psi(\gamma, \beta)|H_C|\psi(\gamma, \beta)\rangle \;=\; \langle +|^{\otimes n}\, U_C^\dagger(\gamma)\, U_B^\dagger(\beta)\, H_C\, U_B(\beta)\, U_C(\gamma)\, |+\rangle^{\otimes n}\,. \tag{19}$$

5.2 Periodicity

For MaxCut on unweighted graphs, the energy landscape exhibits the following periodicity:

- $E(\gamma, \beta)$ is π-periodic in γ, since the cost unitary involves $e^{i\gamma Z_i Z_j}$ and $Z_i Z_j$ has eigenvalues ± 1.
- $E(\gamma, \beta)$ is $\pi/2$-periodic in β (for the standard transverse-field mixer on MaxCut), arising from the symmetry of the R_X rotation and the $\mathbb{Z}_2$ symmetry of the cost function.

It therefore suffices to scan $\gamma \in [0, \pi]$ and $\beta \in [0, \pi/2]$.

5.3 Analytical Expression for $p = 1$

For a d-regular graph with n vertices and $m = |E|$ edges at $p = 1$, the QAOA energy has the closed-form expression (see [1, 2]):

$$E(\gamma, \beta) \;=\; \frac{m}{2} + \frac{1}{2} \sum_{(i,j)\in E} \Big\langle Z_i Z_j \Big\rangle_{\gamma,\beta}, \tag{20}$$

where each $\langle Z_i Z_j\rangle$ depends on the local subgraph around edge (i, j). For the 2-regular cycle C_4:

$$\langle Z_i Z_j\rangle \;=\; \frac{1}{2}\sin(4\beta)\sin(2\gamma)\big[\cos(2\gamma) + \cos^2(2\gamma)\big] - \frac{1}{2}\sin^2(2\beta)\big[1 - \cos^2(2\gamma)\big]. \tag{21}$$

By symmetry all four edges give the same contribution, so:

$$E(\gamma, \beta) \;=\; 2 - 2\langle Z_i Z_j\rangle. \tag{22}$$

5.4 Landscape Visualisations

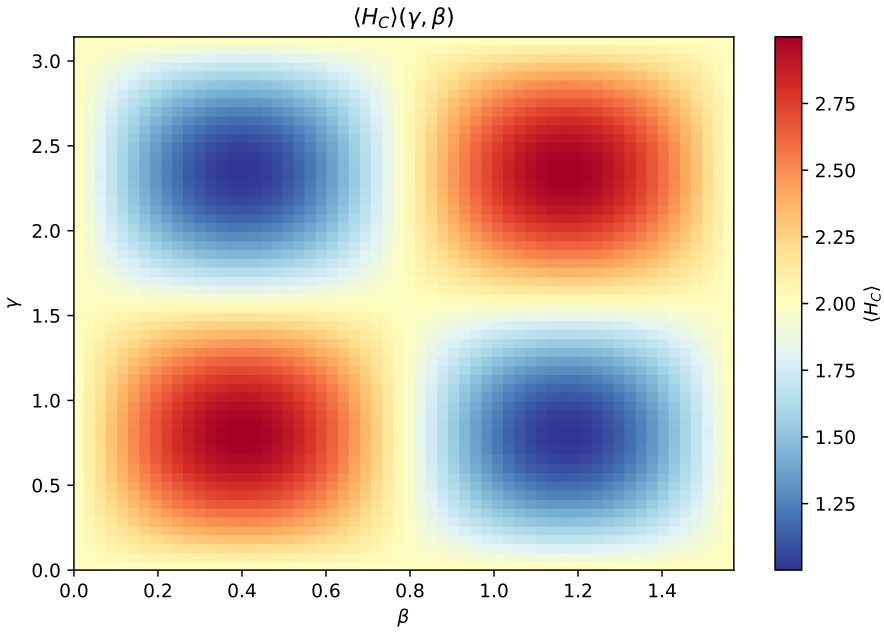

Figure 2: 2D heatmap of the QAOA energy $E(\gamma, \beta)$ for C_4 at $p = 1$. The global maximum (brightest region) occurs near $(\gamma^*, \beta^*) \approx (\pi/4, 5\pi/8)$.

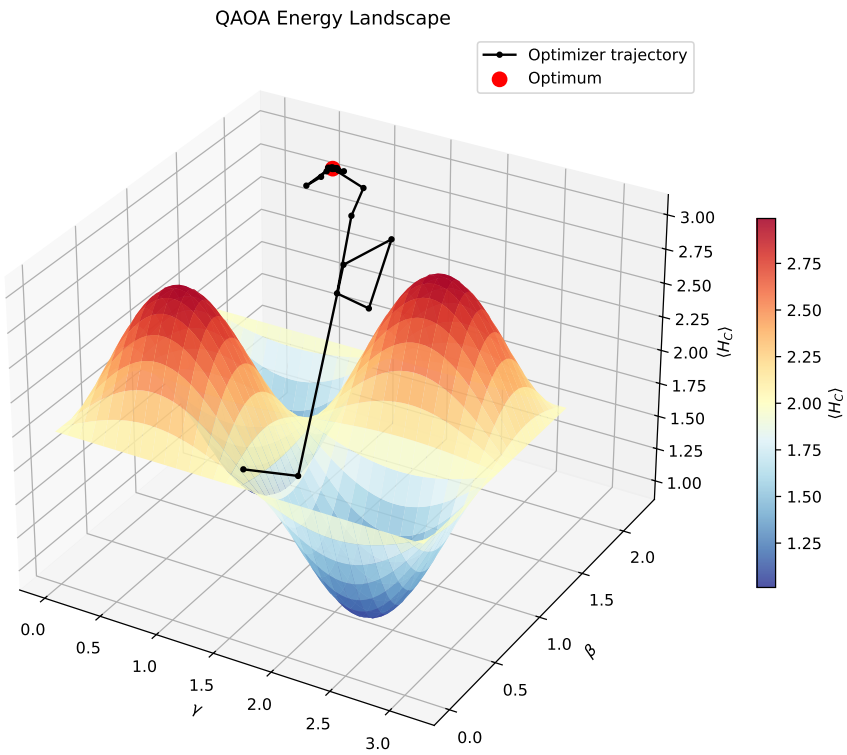

Figure 3: 3D surface plot of the QAOA energy landscape for C_4 at $p = 1$.

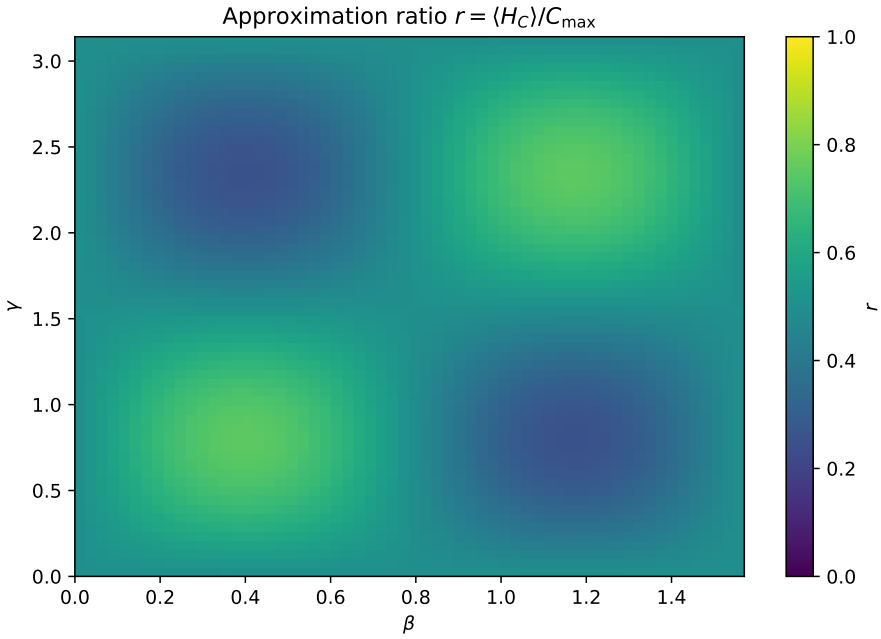

Figure 4: Approximation ratio $r(\gamma, \beta) = E(\gamma, \beta)/C_{\max}$ across the parameter space. The maximum approximation ratio is $r = 0.75$.

5.5 Observed Features

Several features are evident in Figures 2–4:

- **Global maximum**: The energy attains its maximum $E^* = 3.0$ at $(\gamma^*, \beta^*) \approx (\pi/4,\ 5\pi/8)$, yielding an approximation ratio $r = 3/4 = 0.75$.
- **Symmetry**: The landscape is symmetric under $\beta \to \pi/2 - \beta$ (reflection about $\beta = \pi/4$) and exhibits the expected periodicity.
- **Multiple local extrema**: The landscape contains saddle points and local minima, illustrating why gradient-free optimizers or careful initialization are needed.
- **Smooth structure**: The landscape is smooth and differentiable, which is a consequence of the shallow ($p = 1$) circuit and the small graph size.

6 Classical Optimization

6.1 Optimisation Problem

The variational loop seeks to maximise the energy, or equivalently minimise its negation:

$$(\gamma^*, \beta^*) \;=\; \arg\max_{\gamma,\beta} E(\gamma, \beta) \;=\; \arg\min_{\gamma,\beta} \big[-E(\gamma, \beta)\big]. \tag{23}$$

6.2 COBYLA Optimizer

We use the **Constrained Optimization BY Linear Approximations (COBYLA)** algorithm [3], a gradient-free method that:

- Constructs a linear approximation to the objective in a trust region.
- Updates the trust region radius (starting from `rhobeg`) at each iteration.
- Does not require gradient evaluations, making it suitable for noisy quantum objectives.

6.3 Parameter-Shift Rule

Although COBYLA is gradient-free, gradient-based optimizers can exploit the **parameter-shift rule**. For a parameter θ_i appearing in a gate $e^{-i\theta_i G/2}$ where G has eigenvalues ± 1:

$$\frac{\partial E}{\partial \theta_i} = \frac{E(\theta_i + \pi/2) - E(\theta_i - \pi/2)}{2}. \tag{24}$$

This provides exact (not finite-difference) gradients using only two circuit evaluations per parameter, and is an alternative strategy for landscape navigation.

6.4 Convergence

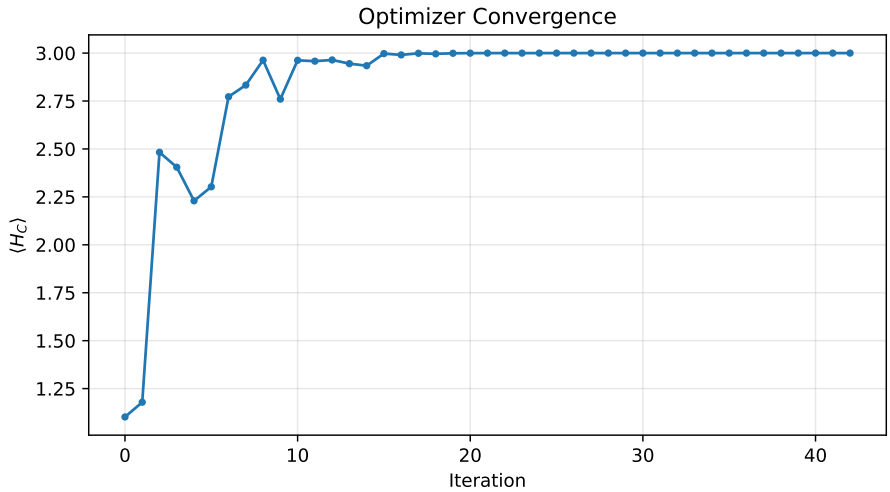

Figure 5: Convergence of the COBYLA optimizer. The algorithm reaches the optimal energy $E^* = 3.0$ within 43 function evaluations.

6.5 Results

The optimizer converged in **43 function evaluations** to:

$$\gamma^* \approx \frac{\pi}{4} \approx 0.7854, \tag{25}$$

$$\beta^* \approx \frac{5\pi}{8} \approx 1.9635, \tag{26}$$

$$E^* = 3.0, \tag{27}$$

$$r^* = \frac{E^*}{C_{\max}} = \frac{3.0}{4.0} = 0.75. \tag{28}$$

6.6 Discussion

The theoretical maximum approximation ratio for $p = 1$ QAOA on the cycle graph C_4 is indeed $r = 0.75$, consistent with the general result that depth-1 QAOA on 2-regular (cycle) graphs achieves:

$$r_{p=1} = \frac{3}{4} \tag{29}$$

for all cycle lengths $n \geq 4$. This bound can be improved by increasing the circuit depth p; for $p \to \infty$, QAOA recovers the exact solution $r \to 1$.

7 Noise Effects and Error Mitigation

7.1 Noise Model

To simulate realistic near-term hardware, we apply a noise model consisting of two channels:

Depolarizing channel. After each gate, a depolarizing channel with error probability $p_{\text{dep}} = 0.01$ is applied:

$$\mathcal{E}_{\text{dep}}(\rho) \;=\; (1 - p_{\text{dep}})\,\rho \;+\; \frac{p_{\text{dep}}}{2^k - 1} \sum_{P \neq I} P\,\rho\,P^\dagger, \tag{30}$$

where the sum runs over all k-qubit Pauli operators (excluding identity) and k is the number of qubits the gate acts on.

Thermal relaxation. We model decoherence with relaxation times:

$$T_1 = 50\ \mu\text{s}, \qquad\qquad T_2 = 70\ \mu\text{s}, \tag{31}$$

and gate durations:

$$t_{1\text{Q}} = 50\ \text{ns} \quad \text{(single-qubit gates)}, \qquad\qquad t_{2\text{Q}} = 300\ \text{ns} \quad \text{(two-qubit gates)}. \tag{32}$$

7.2 Noisy Result

Running the optimal QAOA circuit (γ^*, β^*) under this noise model yields a degraded energy:

$$E_{\text{noisy}} = 2.0. \tag{33}$$

The noise has reduced the approximation ratio from 0.75 to 0.50, a significant degradation that motivates error mitigation.

7.3 Zero-Noise Extrapolation (ZNE)

ZNE [4, 5] is a leading error mitigation technique that:

1. **Amplifies** the noise by a set of scale factors $\{s_1, s_2, \ldots, s_k\}$ with $s_1 = 1$ (no amplification).
2. **Extrapolates** the noisy expectation values to the zero-noise limit $s \to 0$.

Global unitary folding. Noise is amplified by replacing the unitary U with:

$$U \;\longrightarrow\; U\,(U^\dagger U)^n, \tag{34}$$

which implements the circuit $2n + 1$ times (effectively scaling the noise by a factor $s = 2n + 1$) while preserving the ideal unitary action.

Extrapolation. Given expectation values $E(s_1)$, $E(s_2)$, $\ldots$, $E(s_k)$ at different noise scale factors, we estimate the zero-noise value:

- **Linear extrapolation**: Fit $E(s) = a + bs$ and evaluate at $s = 0$:

$$E_{\text{ZNE}}^{\text{lin}} = E(s_1) - s_1\,\frac{E(s_2) - E(s_1)}{s_2 - s_1}. \tag{35}$$

- **Richardson extrapolation**: Use a polynomial or rational fit through all k points to cancel higher-order error terms systematically.

7.4 ZNE Result

Applying ZNE with global unitary folding at scale factors $s \in \{1, 3, 5\}$ and Richardson extrapolation:

$$\boxed{E_{\text{ZNE}} = 3.0,} \tag{36}$$

which fully recovers the noiseless optimal energy. The mitigation has restored the approximation ratio from 0.50 back to 0.75.

7.5 Noise Comparison

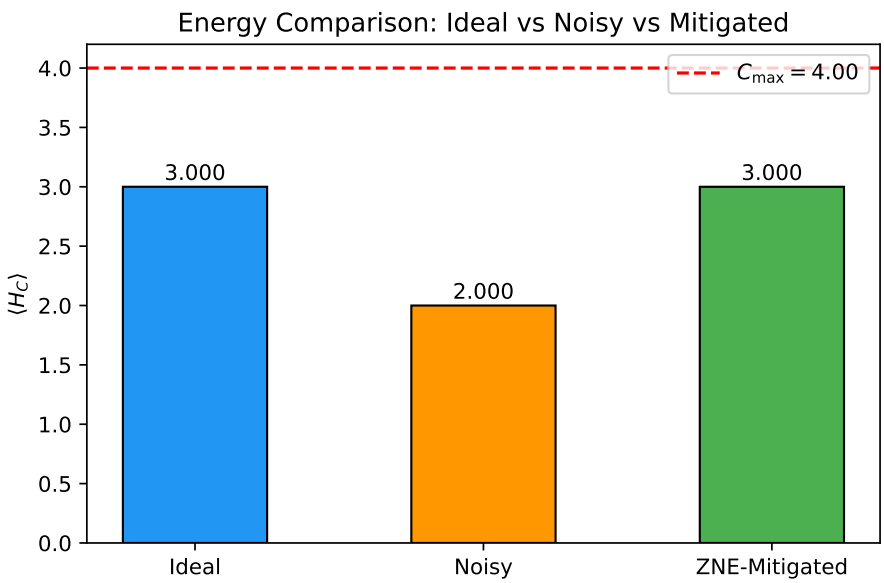

Figure 6: Comparison of energy values: ideal ($E = 3.0$), noisy ($E = 2.0$), and ZNE-mitigated ($E = 3.0$). Zero-noise extrapolation successfully recovers the noiseless result.

8 Conclusion

We have presented a complete, end-to-end mathematical walkthrough of the QAOA Energy Landscape Explorer applied to the MaxCut problem on the 4-node cycle graph C_4. The pipeline consisted of:

1. **Problem formulation**: Defining MaxCut on C_4 and encoding it as an Ising cost function (Section 2).
2. **Hamiltonian construction**: Deriving the diagonal cost Hamiltonian $H_C = 2I - \frac{1}{2}(Z_0Z_1 + Z_1Z_2 + Z_2Z_3 + Z_3Z_0)$ and verifying its eigenvalues against classical cut values (Section 3).
3. **Circuit construction**: Building the $p = 1$ QAOA circuit with IsingZZ cost gates and R_X mixer gates (Section 4).
4. **Energy landscape**: Scanning the two-parameter landscape $E(\gamma, \beta)$, identifying its periodicity and symmetry, and locating the global optimum (Section 5).
5. **Classical optimization**: Using COBYLA to find optimal parameters $\gamma^* \approx \pi/4$, $\beta^* \approx 5\pi/8$, achieving $E^* = 3.0$ and approximation ratio $r = 0.75$ in 43 evaluations (Section 6).
6. **Noise and mitigation**: Simulating depolarizing and thermal relaxation noise ($E_{\text{noisy}} = 2.0$) and recovering the noiseless result via ZNE ($E_{\text{ZNE}} = 3.0$) (Section 7).

This worked example demonstrates that even at the shallowest circuit depth ($p = 1$), QAOA produces a structured, smooth energy landscape with analytically predictable optima, and that zero-noise extrapolation can effectively counteract realistic hardware noise on small instances.

References

[1] E. Farhi, J. Goldstone, and S. Gutmann, "A Quantum Approximate Optimization Algorithm," *arXiv:1411.4028*, 2014.

[2] Z. Wang, S. Hadfield, Z. Jiang, and E. G. Rieffel, "Quantum Approximate Optimization Algorithm for MaxCut: A Fermionic View," *Phys. Rev. A*, vol. 97, p. 022304, 2018.

[3] M. J. D. Powell, "A direct search optimization method that models the objective and constraint functions by linear interpolation," in *Advances in Optimization and Numerical Analysis*, pp. 51–67, 1994.

[4] K. Temme, S. Bravyi, and J. M. Gambetta, "Error mitigation for short-depth quantum circuits," *Phys. Rev. Lett.*, vol. 119, p. 180509, 2017.

[5] Y. Li and S. C. Benjamin, "Efficient variational quantum simulator incorporating active error minimization," *Phys. Rev. X*, vol. 7, p. 021050, 2017.